

```
In [1]: with open("sample_weather.txt", "w") as f:
        f.write("""2026-04-01 Pune 32 22 18 12
2026-04-01 Mumbai 34 25 20 15
2026-04-01 Delhi 36 20 17 10
2026-04-02 Pune 30 21 16 11
2026-04-02 Mumbai 33 26 21 14
2026-04-02 Delhi 37 22 19 13
2026-04-03 Pune 29 20 15 10
2026-04-03 Mumbai 32 25 18 16
2026-04-03 Delhi 38 23 20 12
2026-04-04 Pune 31 21 17 13
2026-04-04 Mumbai 35 27 22 17
2026-04-04 Delhi 39 24 21 14
""")
```

```
In [2]: # -----  
# MAP PHASE  
# -----  
mapped = []  
  
with open("sample_weather.txt", "r") as file:  
    for line in file:  
        parts = line.split()  
  
        # Extract values  
        temp = float(parts[2])  
        dew = float(parts[3])  
        wind = float(parts[5])  
  
        # Emit key-value pairs  
        mapped.append(("temp", temp))  
        mapped.append(("dew", dew))  
        mapped.append(("wind", wind))  
  
print("Mapped Data:")  
print(mapped)  
  
# -----  
# SHUFFLE & SORT PHASE  
# -----  
shuffle = {}  
  
for key, value in mapped:  
    if key not in shuffle:  
        shuffle[key] = []  
    shuffle[key].append(value)  
  
print("\nShuffled Data:")  
print(shuffle)  
  
# -----  
# REDUCE PHASE  
# -----  
result = {}  
  
for key in shuffle:  
    values = shuffle[key]  
    avg = sum(values) / len(values)  
    result[key] = avg  
  
print("\nFinal Averages:")  
print("Average Temperature:", result["temp"])  
print("Average Dew Point:", result["dew"])  
print("Average Wind Speed:", result["wind"])
```

Mapped Data:

```
[('temp', 32.0), ('dew', 22.0), ('wind', 12.0), ('temp', 34.0), ('dew', 25.0), ('wind', 15.0), ('temp', 36.0), ('dew', 20.0), ('wind', 10.0), ('temp', 30.0), ('dew', 21.0), ('wind', 11.0), ('temp', 33.0), ('dew', 26.0), ('wind', 14.0), ('temp', 37.0), ('dew', 22.0), ('wind', 13.0), ('temp', 29.0), ('dew', 20.0), ('wind', 10.0), ('temp', 32.0), ('dew', 25.0), ('wind', 16.0), ('temp', 38.0), ('dew', 23.0), ('wind', 12.0), ('temp', 31.0), ('dew', 21.0), ('wind', 13.0), ('temp', 35.0), ('dew', 27.0), ('wind', 17.0), ('temp', 39.0), ('dew', 24.0), ('wind', 14.0)]
```

Shuffled Data:

```
{'temp': [32.0, 34.0, 36.0, 30.0, 33.0, 37.0, 29.0, 32.0, 38.0, 31.0, 35.0, 39.0], 'dew': [22.0, 25.0, 20.0, 21.0, 26.0, 22.0, 20.0, 25.0, 23.0, 21.0, 27.0, 24.0], 'wind': [12.0, 15.0, 10.0, 11.0, 14.0, 13.0, 10.0, 16.0, 12.0, 13.0, 17.0, 14.0]}
```

Final Averages:

Average Temperature: 33.833333333333336

Average Dew Point: 23.0

Average Wind Speed: 13.083333333333334

In []: